

1. Administrative & Study Identification

Full Study Title: Study of Intratumoral (IT) Ulevostinag (MK-1454) in Combination With Intravenous (IV) Pembrolizumab (MK-3475) Compared to IV Pembrolizumab Alone as the First Line Treatment of Metastatic or Unresectable, Recurrent Head and Neck Squamous Cell Carcinoma (HNSCC) (MK-1454-002) **Short Title/ Acronym:** MK-1454-002 **Protocol Number:** MK-1454-002 **NCT Number:** NCT04220866 **Posting ID:** 936743557073299344 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Ulevostinag (MK-1454) [STING agonist] and Pembrolizumab (Trade Name: Keytruda, ATC Code: L01FF02) [anti-PD-1] **Phase of Development:** Phase II **Trial Status:** Completed **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of intratumoral (IT) ulevostinag PLUS pembrolizumab compared to pembrolizumab alone as a first-line treatment for adults with metastatic or unresectable, recurrent head and neck squamous cell carcinoma (HNSCC). The primary hypothesis is that the combination results in a superior Objective Response Rate (ORR) per RECIST 1.1 in participants with a PD-L1 Combined Positive Score (CPS) ≥ 1 and CPS ≥ 20 . **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Duration of Response (DOR), Overall Survival (OS), and overall safety and tolerability of the combination regimen.

2.2 Background & Rationale To address the need for enhanced innate and adaptive immune responses in treating HNSCC. Activation of the STING (stimulator of interferon genes) pathway by intratumoral injection of ulevostinag modulates the tumor microenvironment by inducing type I IFNs and proinflammatory cytokines. Adding a PD-1 inhibitor (pembrolizumab) is hypothesized to maintain the functional capacity of activated T cells, resulting in improved tumor shrinkage in both injected and non-injected lesions compared to immunotherapy alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab monotherapy) **Blinding:** Open-label (with Blinded Independent Central Review [BICR] for efficacy endpoints) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 18 participants
Number of Sites: Multicenter / Global **Estimated Study Start:** 04-Mar-2020 **Estimated Primary Completion:** 30-Sep-2022 (Actual) **Trial Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of metastatic or unresectable, recurrent HNSCC. 3. At least one measurable lesion amenable to intratumoral injection. 4. Tumor expressing PD-L1 with a CPS ≥ 1 .

4.2 Exclusion Criteria 1. Prior systemic therapy for metastatic or unresectable recurrent HNSCC (must be first-line treatment). 2. Prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent. 3. Prior treatment with a STING agonist. 4. Active autoimmune disease or a history of (non-infectious) pneumonitis requiring systemic steroids.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Combination Arm:** Ulevostinag 540 μ g via intratumoral (IT) injection on Day 1 of every week for two 3-week cycles (Cycles 1-2), then on Day 1 of each 3-week cycle for up to 33 cycles (Total 35 cycles). This is given PLUS Pembrolizumab (Keytruda) 200 mg via IV infusion on Day 1 of each 3-week cycle. * **Monotherapy Arm:** Pembrolizumab (Keytruda) 200 mg via IV infusion on Day 1 of each 3-week cycle. **Route of Administration:** Intratumoral (IT) for Ulevostinag; Intravenous (IV) for Pembrolizumab. **Duration of Treatment:** Up to 35 cycles (approximately 2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology patients. **Prohibited:** Systemic immunosuppressive or immunomodulatory medications (e.g., high-dose corticosteroids, unless for managing immune-mediated adverse events), other investigational therapies, and live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Imaging, PD-L1 Testing, Biopsies, Safety Labs
Baseline/Cycle 1	Day 1	Randomization, First IV infusion ($\pm$ IT injection), PK/Biomarker draws
Treatment	Cycles 1-35	IV/IT dosing every 3 weeks (Weekly IT injections during Cycles 1-2 for combo arm). Routine safety labs and AE assessments.
End of Study	Up to 2 years	Treatment discontinuation, Safety follow-up (30/90 days), Survival tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), defined as the percentage of participants with a Complete Response (CR) or Partial Response (PR) per RECIST 1.1.

Measurement Method: Assessed by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) per RECIST 1.1 by BICR. * Duration of Response (DOR) per RECIST 1.1 by BICR. * Overall Survival (OS). * Safety and tolerability (e.g., incidence of AEs such as pyrexia) and Dose-Limiting Toxicities.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Graded using NCI Common Terminology Criteria for Adverse Events (CTCAE) version 5.0.

Specific monitoring for immune-mediated AEs, pyrexia, and injection-site reactions. **Serious Adverse Events (SAEs):** Standard reporting timelines applied for treatment-emergent and immune-related severe toxicities. **Data Monitoring Committee (DMC):** Utilized for periodic safety review and interim analysis evaluations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for primary efficacy (ORR, PFS, OS); Safety Set for adverse events.

Sample Size Justification: Planned target enrollment was 18 participants. **Handling of Missing Data:** Product-limit (Kaplan-Meier) method used for time-to-event endpoints (PFS, DOR, OS) with standard censoring rules. Exact method for binomial data utilized for ORR 95% Confidence Intervals.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine site visits and centralized data monitoring to ensure protocol adherence and safety. **Audit Trail:** Secure electronic Case Report Forms (eCRF) utilizing strict data locking and query resolution workflows. Blinded Independent Central Review (BICR) used to objectively verify tumor responses.

11. Study Identifiers & Governance

Primary ID: {{MK-1454-002}} **Posting ID:** {{936743557073299344}}
Registry Number: {{NCT04220866}} **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC (Merck)}} **Study Title:** {{MK-1454-002: Ulevostinag + Pembrolizumab vs. Pembrolizumab in HNSCC}} **Official Regulatory Title:** {{A Phase 2 Study in First Line Metastatic or Unresectable, Recurrent Head and Neck Squamous Cell Carcinoma to Evaluate Intratumoral MK-1454 in Combination With IV Pembrolizumab vs IV Pembrolizumab Monotherapy}}

12. Clinical Framework (Oncology / HNSCC Specific)

Primary Condition: {{Metastatic or Unresectable, Recurrent HNSCC}} **Preferred Disease Name:** {{Carcinoma}} **Diagnosis Threshold:** {{PD-L1 CPS $\geq$ 1, measurable disease per RECIST 1.1}} **Medical Methodology:** {{Intratumoral STING Agonist combined with IV anti-PD-1 Immunotherapy}} **Optimization Target:** {{Enhanced anti-tumor activity and Objective Response Rate (ORR)}}

13. Study Procedures & Methodology

Management Pathway: {{First-line Oncology Immunotherapy Pathway}} **Education Protocol (HCP):** {{Image-guided / Palpation-guided Intratumoral Injection Training}} **Education Protocol (Patient):** {{Oncology Infusion and Injection Site Care Education}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for radiographic response}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 2 years of treatment + survival follow-up}} **Onsite Visit Cadence:** {{Day 1 of every 3-week cycle; weekly during Cycles 1 & 2 for the combination arm}} **Remote Monitoring Frequency:** {{Standard survival follow-up contacts post-treatment}} **Checklist Review Interval:** {{Tumor imaging typically every 9 weeks for the first year, then every 12 weeks}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				